

Philip Hargreaves

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Summary

MSc Computer Science student at University College London with a strong engineering background and hands-on software development experience. Actively seeking full-time software engineering roles.

Education

University College London	Sept 2025 – Sept 2026
MSc Computer Science	Expected Distinction

- **Modules:** Data Structures and Algorithms, Computer Architecture, Operating Systems, Research Computing with C++, Object-Oriented Programming, Machine Learning, Database Fundamentals

University of Nottingham	Sept 2018 – Sept 2022
MEng Mechanical Engineering	First Class Honours

Experience

Mechanical Design Engineer , Rolls-Royce	May 2023 – Sept 2025
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- Developed a Python-based design toolkit using Model-View-Presenter architecture and Tkinter GUI, integrating NumPy, Pandas, and Matplotlib, reducing preliminary vessel sizing time by 30%
- Achieved a 40% life extension for a critical nuclear component by developing refined brittle fracture methodologies, securing ONR regulatory approval and >£50k in recurring operational savings
- Automated FEA post-processing using Python, decreasing manual effort and ensuring data consistency
- Validated sub-contractor deliverables as technical reviewer, verifying engineering calculations and models to ensure 100% compliance with safety case requirements

Projects

Full-Stack Auction Platform	PHP, MySQL, JavaScript, Apache, Docker
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- Created custom RESTful MVC framework routing 27 endpoints using dependency injection, repository pattern and a normalised, 16-table MySQL schema supporting recursive category filtering
- Developed transaction-safe bidding engine with outbid detection, reserve-price checks and winner resolution
- Designed weighted recommendation system using collaborative filtering, in-memory caching and SQL CTEs

Feed Ingestion Service	Go, PostgreSQL, SQLC, Goose
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- Built Go service that ingests RSS feeds into PostgreSQL, normalises entries and serves personalised streams
- Implemented concurrent ingestion using worker pools, batch scheduling and domain-level rate limiting
- Developed a process supervisor with exponential backoff to auto-recover from network or panic failures

AI Coding Agent	Python, Gemini API, Pytest
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- Developed an autonomous CLI agent that analyses and debugs codebases via Gemini API function calling, supporting multi-turn context, streaming responses and automated tool orchestration
- Engineered a sandboxed agentic loop with file I/O, shell command validation and diff-based patching

Debugging Memory Allocator	C, Linux, GDB, Docker
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- Built a debugging memory allocator in C that intercepts heap calls to detect double-frees, boundary write overflows and leaks via per-block metadata and 16-byte aligned payloads
- Implemented a heavy hitter profiler using frequency estimation to rank allocation sites by byte volume

Skills

Programming Languages: Python, Go, Java, C, C++, JavaScript/TypeScript, SQL

Frameworks / Libraries: net/http, chi, SQLC, Goose, FastAPI, Pytest, React, Prisma, Next.js, Node.js

Tools / Technologies: Git, Docker, AWS (Certified Cloud Practitioner), Linux, MySQL, PostgreSQL, Apache